

The Next Ravi Varma

Generating Ravi Varma style paintings using Stable Diffusion + LoRA

Progress Presentation (Milestones 1 and 2)

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(generated by our model)

Introduction

- Raja Ravi Varma (1848-1906) was a famous Indian painter known for realistic oil paintings of Indian mythology and royal portraits.
- Inspired by "The Next Rembrandt" project, which generated a new painting in Rembrandt's style.
- Our goal: give a text description of a scene and generate a new image in Ravi Varma's style.
- We use Stable Diffusion 1.5 and fine-tune it with LoRA on Ravi Varma's paintings.



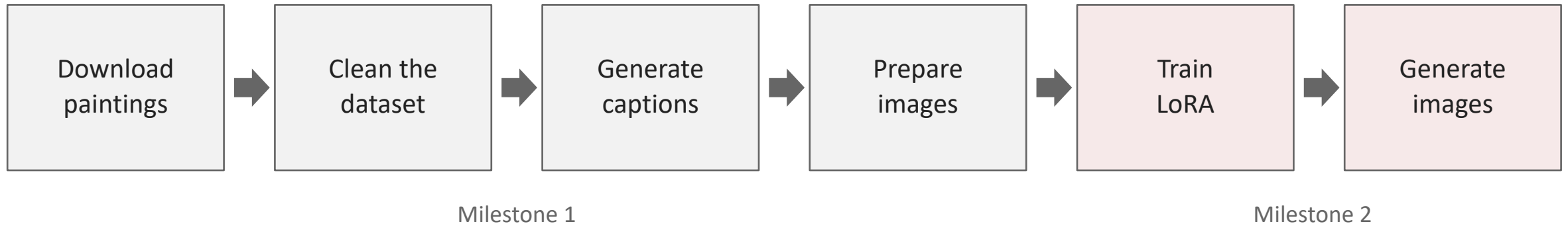
Hamsa Damayanti by Raja Ravi Varma

Project Milestones and Status

Milestone	Work	Status
1	Dataset collection, cleaning and captioning	Done
2	LoRA fine-tuning of Stable Diffusion	Done
3	Prompt expansion using an LLM	Not started
4	ControlNet (pose control), evaluation, web UI	Not started

Overall progress: about 50%

Overall Pipeline



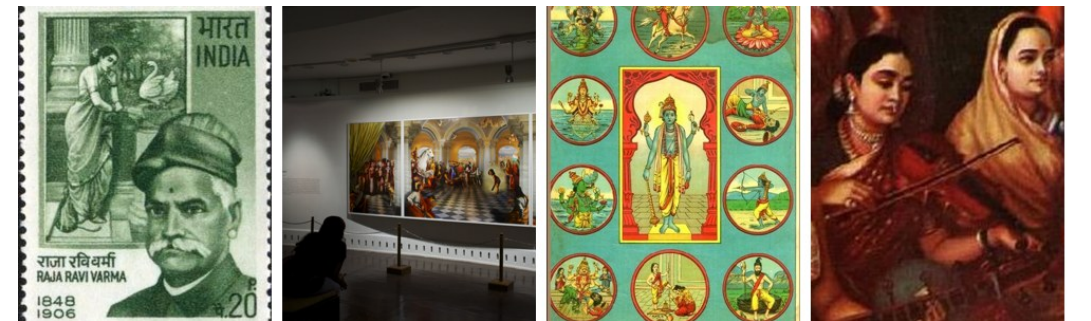
- Steps 1-4 run on my laptop (MacBook), training runs on Google Colab (T4 GPU).
- Everything is controlled by config files and scripts, so the results can be reproduced.

Milestone 1: Dataset

- Downloaded 200 Ravi Varma paintings from Wikimedia Commons (most are public domain).
- Removed 66 images:
 - 26 were low resolution (upscaled from very small originals)
 - 33 were duplicates of the same painting
 - 7 were not paintings (stamps, museum photos, grids)
- Final: 134 images, 122 for training and 12 kept aside for testing.
- Images resized without cutting off heads/feet (different sizes for portrait and landscape).



Some training images



Examples of removed images

Milestone 1: Image Captioning



- Each image needs a text caption for training.
- Used BLIP (an image captioning model) to describe the image and BLIP-VQA to answer fixed questions (clothing, jewellery, pose, setting, etc.).
- Wrong answers like "phone" were filtered out.
- Every caption starts with a trigger word <rvvarma> so the model links it to the style.

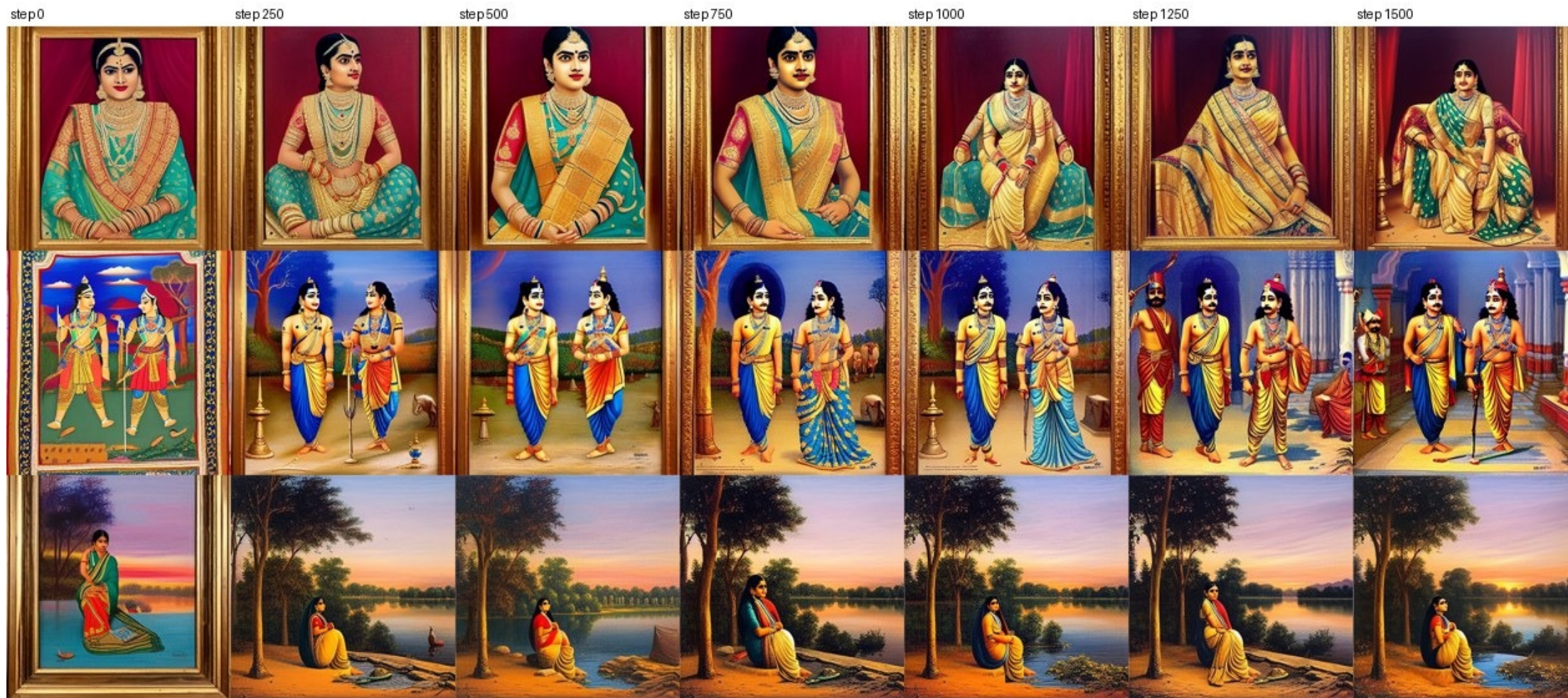
Example caption: <rvvarma>, Hamsa Damayanti, a woman in a red sari feeding a white swan, wearing red dress, necklace jewelry, sitting, calm expression ...

Milestone 2: LoRA Training

- LoRA trains only a small number of extra weights; the original Stable Diffusion model is not changed.
- Only 3.2 million parameters are trained (0.37% of the model).
- Trained on Google Colab with a T4 GPU, took about 1 hour for 1500 steps.
- Saved a checkpoint every 250 steps and generated test images to track progress.

Parameter	Value
Base model	Stable Diffusion 1.5
LoRA rank / alpha	16 / 16
Learning rate	0.0001
Batch size	1 (gradient accumulation 4)
Training steps	1500
Precision	fp16

How the Output Changed During Training



Same prompts and seeds, from step 0 (no training) to step 1500

Results: Without vs With Our LoRA

Without



With LoRA



Royal woman in a palace

Without



With LoRA



Royal portrait

Without



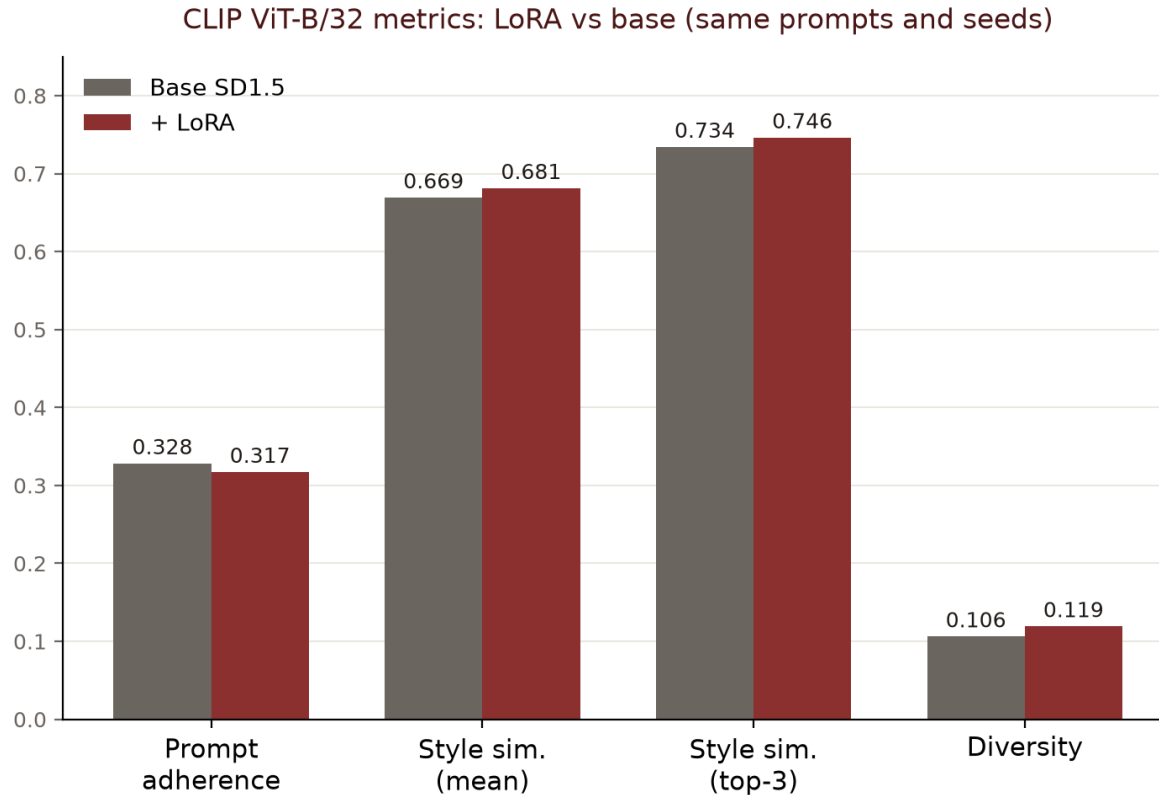
With LoRA



Woman beside a window

- With LoRA the images look more like Ravi Varma's style: full-figure poses, pillars, carpets, darker painted backgrounds.
- The base model added picture frames in 5 of 15 images; our model added none.

Evaluation and Problems



CLIP scores for 15 image pairs

What we found

- CLIP style score increased only slightly (0.669 to 0.681), even though the images look clearly different.
- No image was a copy of a training image.

Problems

- Some women's faces get a dark shadow above the lip.
- Small signature-like marks appear in the corners.
- Group scenes still look like posters.

Challenges Faced

- Many images in the dataset were duplicates or had been upscaled, so we had to check every image.
- The original Stable Diffusion 1.5 model was removed from Hugging Face, so we switched to the official mirror.
- Version conflicts on Google Colab (xformers, bitsandbytes) broke training at first.
- On the Mac, images came out completely black because of an fp16 issue; fixed by changing a setting.
- Free Colab has limited GPU time, so we added checkpoints to resume training if it disconnects.

Future Work

- Fix the face shadow and signature problems by improving the captions.
- Milestone 3: use an LLM to turn a story description into a detailed prompt (e.g. "Krishna and Arjuna at Kurukshetra").
- Milestone 4: add ControlNet so the user can control the pose of the people.
- Build a simple web interface (Gradio) and do a better evaluation, including human ratings.



Current output for a Krishna and Arjuna prompt

Thank You

Questions?

Code and results: github.com/f20240684-cmd/the-next-ravi-varma-v2